

Ex Distribution and density function of sum of R.V. :-

Sum of two Random variables :-

Let w be a random variable equal to the sum of two independent random variables x and y

$$w = x + y \quad \text{then}$$

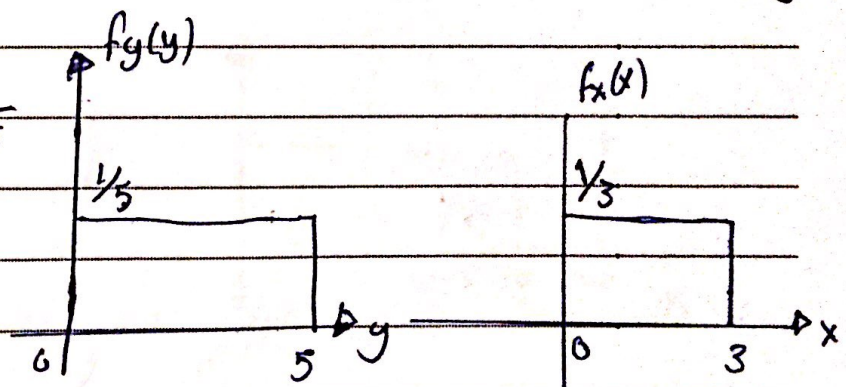
$$f_w(w) = f_x(x) * f_y(y) = \int_{-\infty}^{\infty} f_x(w-y) \cdot f_y(y) \cdot dy$$

$$\text{or} \quad = \int_{-\infty}^{\infty} f_y(w-x) f_x(x) \cdot dx$$

Example :- $f_{xy}(x,y) = \begin{cases} \frac{1}{15} & 0 \leq x \leq 3 \\ & 0 \leq y \leq 5 \\ 0 & \text{else} \end{cases}$ find $f_w(w)$ where $w = x + y$

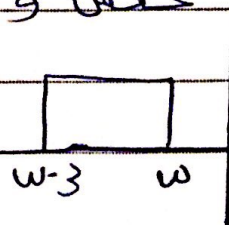
$$f_x(x) = \frac{1}{3}$$

$$f_y(y) = \frac{1}{5}$$



(1) x و y متغيران عشوائيان مستقلان

متغير w هو مجموع x و y



(2) w هو مجموع x و y

①

$w-3$

w

0

5

المفترقات

① $0 < w < 3$

$$f_w(w) = \int_0^w f_y(w-x) f_x(x) dx$$

$$= \int_0^w \frac{1}{15} dx = w/15$$

$w-3$

0

$w/3$

5

② $3 < w < 5$

$$f_w(w) = \int_{w-3}^w \frac{1}{15} dx = \frac{1}{5}$$

$w-3$

3

w

5

③ $5 < w < 8$

$$f_w(w) = \int_{w-3}^5 \frac{1}{15} dx =$$

$$= \frac{(8-w)}{15}$$

$w-3$

5

w

8

$f_w(w) =$

$w/15$

$0 < w < 3$

$1/5$

$3 < w < 5$

$8-w$

$5 < w < 8$

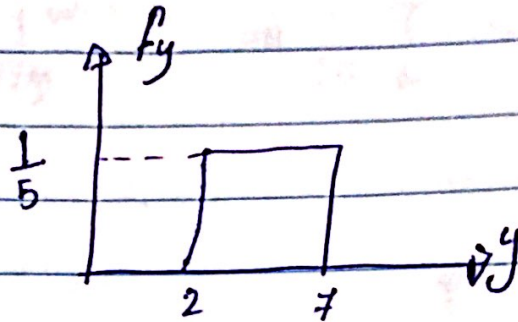
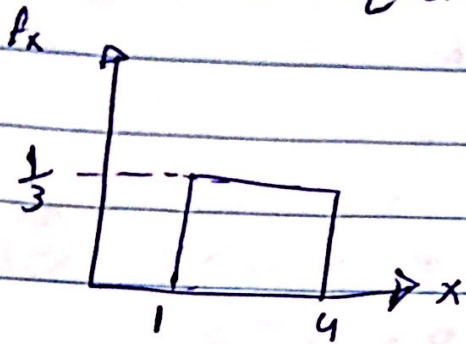
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f(2) (2)

Example 2: x, y random value if $w = x + y$
and given that

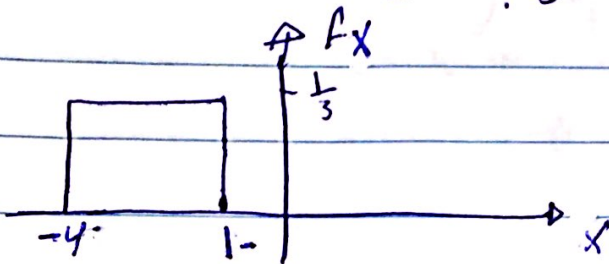
$$f_x(x) = \frac{1}{3} [u(x-1) - u(x-4)]$$

$$f_y = 0.5 [u(y-2) - u(y-7)]$$

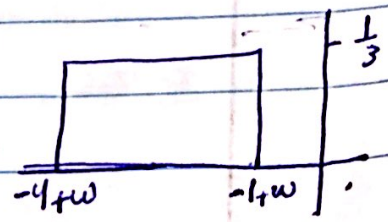


①

بما، واه وبعده



shift
= w بغير



②

الفترة



①

$$w - 1 < 2 \Rightarrow w < 3$$

لا يوجد تداخل
 $f_w(w) = \text{zero}$

②

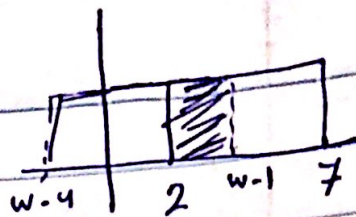
$$1 < w - 1 < 2 \Rightarrow 3 < w < 6$$

$$w - 4 < 2$$

$$3 < w < 6$$

$$\int_2^{w-1} \frac{1}{3} \cdot \frac{1}{5} dy$$

$$\Rightarrow \frac{1}{15} y \Big|_2^{w-1} \Rightarrow \frac{1}{15} [w-3]$$

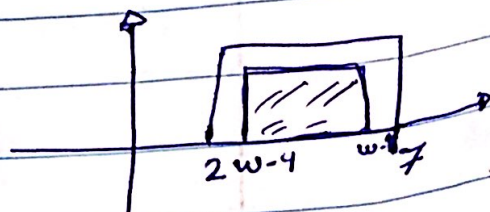


③

$$6 < w < 8$$

$$f_w(w) = \int_{w-4}^{w-1} \frac{1}{3} \cdot \frac{1}{5} dy$$

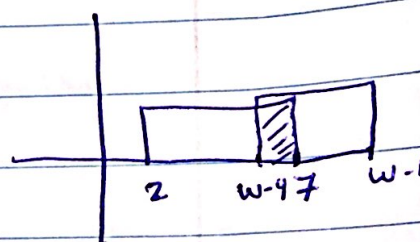
$$\Rightarrow \frac{1}{15} (y) \Big|_{w-4}^{w-1} \Rightarrow \frac{3}{15}$$



④ $8 < w < 11$

$$\int_{w-4}^7 \frac{1}{5} \cdot \frac{1}{3} dy$$

$$\frac{1}{15} y \Big|_{w-4}^7 \Rightarrow \frac{1}{15} (7 - w + 4)$$



⑤ $w > 11$

$$f_w(w) = 0$$

(5)

" Discrete "

Example x and y random variable given by

$$f_x(x) = 0.1\delta(x-1) + 0.2\delta(x-2) + 0.4\delta(x-3) + 0.3\delta(x-4)$$

$$f_y(y) = 0.4\delta(y-5) + 0.5\delta(y-6) + 0.1\delta(y-7)$$

Find f_w if $w = x + y$ and if x, y are Independent ??

$$f_w(w) = \int_{-\infty}^{\infty} f_y(y) f_x(w-y) dy$$

$$\int_{-\infty}^{\infty} [0.4\delta(y-5) + 0.5\delta(y-6) + 0.1\delta(y-7)]$$

$$\times [0.1\delta(w-y-1) + 0.2\delta(w-y-2) + 0.4\delta(w-y-3) + 0.3\delta(w-y-4)]$$

بجواب بیش از این قیّم ی وجوده (5, 6, 7)

$$\Rightarrow \int \delta(w-y-x)$$

prob ۱۱۱۱۱۱

$$\Rightarrow f_w(w) = 0.04\delta(w-6) + 0.08\delta(w-7) + 0.16\delta(w-8)$$

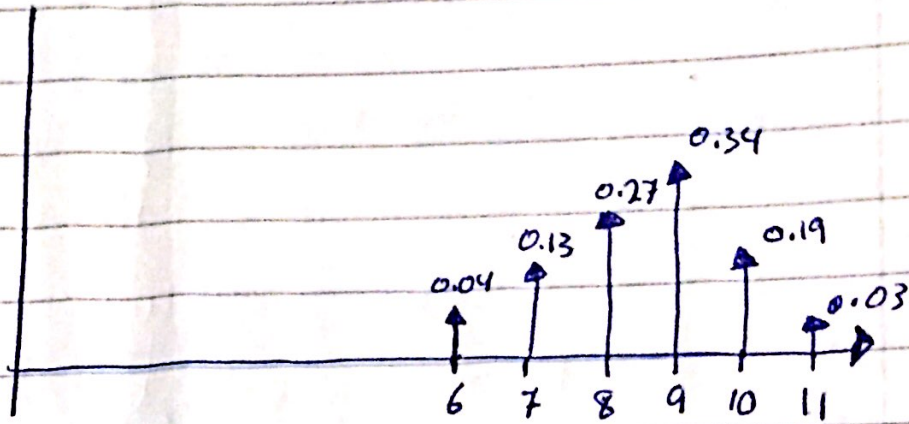
$$+ 0.12\delta(w-9) + 0.05\delta(w-7) + 0.1\delta(w-8)$$

$$+ 0.20\delta(w-9) + 0.15\delta(w-10) + 0.01\delta(w-8)$$

$$+ 0.02\delta(w-9) + 0.04\delta(w-10) + 0.03\delta(w-11)$$

نجمع الحدود المتشابهة

$$f_w(w) = 0.04\delta(w-6) + 0.13\delta(w-7) + 0.27\delta(w-8) \\ + 0.34\delta(w-9) + 0.19\delta(w-10) + 0.03\delta(w-11)$$



Chapter '5'

6. Operations on Multiple Random Variables ::

• Expected value of a function ::

$$E(x^n y^k) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x^n y^k f_{xy}(x, y) dx dy \rightarrow \text{R.V. continuous}$$

$$\rightarrow \sum \sum x^n y^k p(x_i, y_j) \rightarrow \text{R.V. Discrete}$$

• joint moments about the origin ::

denoted by m_{nk} and are defined by ::

$$m_{nk} = E[x^n y^k] = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x^n y^k f_{xy}(x, y) dx dy$$

$$m_{01} = E(x^0 y^1) = E(y) = \bar{y} = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x^0 y \cdot f_{xy}(x, y) dx dy$$

$$\Rightarrow E(y) = \int_{-\infty}^{+\infty} y f_y(y) \cdot dy = \bar{y}$$

$$m_{10} = E(x^1 y^0) = E(x) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x f_{xy}(x, y) dx dy$$

$$\Rightarrow E(x) = \int_{-\infty}^{+\infty} x \cdot f_x(x) \cdot dx = \bar{x}$$

where m_{01}, m_{10} first order moments

$$m_{11} = E(x'y') = \int \int_{-\infty}^{\infty} xy f_{xy}(x,y) = R_{xy}$$

$R_{xy} \equiv$ Correlation of x and y

note: the sum of $(n+k)$ is called order of the moment

$m_{02}, m_{20}, m_{11} \Rightarrow$ Second order moments

$$R_{xy} = E(xy) = \int \int_{-\infty}^{\infty} xy f_{xy}(x,y) dx dy$$

- if $R_{xy} = E(x)E(y) = E(xy)$ then x, y uncorrelated and independent
- if $R_{xy} = 0$ then x, y are called orthogonal

• Joint Central moment :

$$M_{nk} = E[(x-\bar{x})^n (y-\bar{y})^k] = \int \int_{-\infty}^{\infty} (x-\bar{x})^n (y-\bar{y})^k f_{xy}(x,y) dx dy$$

$$M_{01} = M_{10} = 0$$

$$M_{02} = E[(y-\bar{y})^2] = \sigma_y^2$$

$$M_{20} = E[(x-\bar{x})^2] = \sigma_x^2$$

→ Second order central moments

$$M_{11} = C_{xy} = E[(x - \bar{x})(y - \bar{y})] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \bar{x})(y - \bar{y}) f_{xy}(x, y) dx dy$$

$C_{xy} \equiv$ Covariance of x and y

$$C_{xy} = R_{xy} - E(x)E(y) = R_{xy} - \bar{x}\bar{y}$$

→ if x, y uncorrelated or independent $E(x)E(y) = R_{xy}$
then $C_{xy} = 0$

→ if x, y are orthogonal ($R_{xy} = 0$) then $C_{xy} = -\bar{x}\bar{y}$

→ if $[E(x) \text{ or } E(y)]$ zero mean value and x, y orthogonal
then $C_{xy} = 0$

- the normalized second order moment "correlation coefficient"

$$\rho = \frac{C_{xy}}{\sigma_x \sigma_y}, \quad -1 \leq \rho \leq 1$$

$$\rho = E \left[\frac{(x - \bar{x})}{\sigma_x} \frac{(y - \bar{y})}{\sigma_y} \right]$$

Example 20 Let X be a random variable that has a mean value $\bar{X} = E(X) = 3$ and variance $(\sigma_x^2) = 2$ and another random variable Y be defined by $Y = -6X + 22$

Find $E(X^2)$, σ_y^2 , R_{xy} , C_{xy} , ρ_{xy}

$$\bullet E(X^2) = m_{20} = \sigma^2 + \bar{X}^2 \Rightarrow 2 + 3^2 = 11$$

$$\bullet \sigma_y^2 = \sigma^2 \cdot \sigma_x^2 = 36 \times 2 = 72$$

$$\bullet \rho_{xy} = m_{11} = E(XY) = E[-6X^2 + 22X] = -6E[X^2] + 22\bar{X} \\ = -6 \times 11 + 22 \times 3 \\ = 0$$

the X, Y are orthogonal

$$\bullet C_{xy} = R_{xy} - E(X) \cdot E(Y)$$

$$= 0 - 3 \times E(Y) \Rightarrow E(Y) = E(-6X + 22)$$

$$= 0 - 3 \times 4$$

$$= -6\bar{X} + 22 = +4$$

$$= -12$$

$$\bullet R_{xy} = \frac{C_{xy}}{\sigma_x \sigma_y} = \frac{-12}{\sqrt{2 \times 72}}$$

Example 20 Random variables X and Y have density function

$$f_{xy}(x, y) = \begin{cases} \frac{1}{24} & 0 < x < 6, 0 < y < 6 \\ 0 & \text{elsewhere} \end{cases}$$

find m_{10} , m_{01} , R_{xy} , m_{21}

$$m_{10} = \bar{x} = \int_0^6 \int_0^6 y^0 x^1 \cdot \frac{1}{24} dx dy \Rightarrow \frac{9}{2}$$

$$m_{01} = \bar{y} = \int_0^6 \int_0^6 y^1 x^0 \cdot \frac{1}{24} dx dy \Rightarrow \frac{9}{2}$$

$$R_{xy} = E(xy) = \int_0^6 \int_0^6 xy \cdot \frac{1}{24} dx dy \Rightarrow 13.5$$

$$m_{21} = \int_0^6 \int_0^6 x^2 y \cdot \frac{1}{24} dx dy \Rightarrow 54$$

Example: the statistical independence Random variable (x,y) if $\bar{x}=2$, $\bar{y}=4$, $\bar{x}^2=8$, $\bar{y}^2=25$ find

- ① the mean value ($\bar{w}$) where $w=3x-y$
- ② Variance of w and second order of w

$$1) w=3x-y \Rightarrow E(w) = 3E(x) - E(y) \Rightarrow 3\bar{x} - \bar{y} = 3 \times 2 - 4 = 2$$

$$2) E(w^2) = E(3x-y)^2 = E(9x^2 - 6xy + y^2) = 9E(x^2) - 6E(xy) + E(y^2)$$

where

$$E(x^2)=8, E(y^2)=25 \quad \text{independence}$$

$$\text{then } E(xy) = E(y) \cdot E(x)$$

$$2 \times 4 = 8$$

statistical indep. is

(12)

$$\bar{w}^2 = 9 \times 8 - 6(8) + 25 = 49$$

$$6w^2 = m_2 - m_1^2 = 49 - 4 = 45$$

Example 2: $f_{xy}(x,y) = 0.15 \delta(x+1) \delta(y) + 0.1 \delta(x) \delta(y) + 0.1 \delta(x) \delta(y-2) + 0.4 \delta(x-1) \delta(y+4) + 0.2 \delta(x-1) \delta(y-1) + 0.05 \delta(x-1) \delta(y-3)$

Find $E(x)$, $E(y)$, $E(x^2)$, $6x^2$, $6y^2$, R_{xy} , C_{xy}

$$\bullet E(x) = 0.15(-1) + 0.1(0) + 0.1(0) + 0.4(1) + 0.2(1) + 0.05(1) = 0.5$$

$$\bullet E(y) = 0.15(0) + 0.1(0) + 0.1(2) + 0.4(-4) + 0.2(1) + 0.05(3) = -1.05$$

$$\bullet E(x^2) = 0.15(-1)^2 + 0.1(0)^2 + 0.1(0)^2 + 0.4(1)^2 + 0.2(1)^2 + 0.05(1)^2 = 0.8$$

$$6x^2 = E(x^2) - E(x)^2 = 0.55$$

$$m_2 - m_1^2$$

$$6y^2 = E(y^2) - E(y)^2 \rightarrow E(y^2) = 7.45$$

$$= 6.35 \quad E(y) = -1.05$$

$$m_{11} = R_{xy} = 0.15(-1)(0) + 0.1(0)(0) + 0.1(0)(2) + 0.4(1)(-4) + 0.2(1)(1) + 0.05(1)(3) = -1.15$$

$$C_{xy} = R_{xy} - E(x)E(y) = -1.15 - (0.5)(-1.05) = -0.575$$

$$m_{21} = 0.15(-1)^2(0) + 0.1(0)^2(0) + 0.1(0)^2(2) + 0.4(1)^2(-4) + 0.2(1)^2(1) + 0.05(1)^2(3) = -1.15$$

(13)

example : if $\bar{x}=1$ $\sigma_x^2=4$ $\rho_{xy}=0.4$
 $\bar{y}=2$ $\sigma_y^2=1$ if $w=x+3y$
 $v=-x+2y$

find $\bar{w}$, $\bar{v}$, σ_w^2 , σ_v^2 , C_{vw} , R_{vw} , ρ_{vw}

① $\bar{w} = \bar{x} + 3\bar{y} = 7$ ② $\bar{v} = -\bar{x} + 2\bar{y} = 3$

③ $\bar{w}^2 = E(x^2 + 6xy + 9y^2) = E(x^2) + 6E(xy) + 9E(y^2)$

where

$E(x^2) = \sigma_x^2 + (m_1)^2 = 5$ $5 + 6(2.8) + 9 \times 5$

$E(y) = \sigma_y^2 + (m_1)^2 = 5$ $= 66.8$

$R_{xy} = E(xy) = 2.8$

$\rho_{xy} = \frac{C_{xy}}{\sqrt{\sigma_x^2 \sigma_y^2}} = C_{xy} = 0.4 \times 1 \times 2 = 0.8$

$C_{xy} = R_{xy} - E(x)E(y) \Rightarrow R_{xy} = 0.8 + 1 \times 2 = 2.8$

④ $\sigma_w^2 = m_2 - (m_1)^2 = 66.8 - 49 = 17.8$

⑤ $E(v^2) = E(x^2) + 4(y^2) - 4E(xy) = 5 + 20 - 4(2.8) = 13.8$

$\sigma_v^2 = m_2 - (m_1)^2 = 13.8 - 9 = 4.8$

⑥ $R_{vw} = E(vw) = E[(x+3y)(-x+2y)]$

$= -E(x^2) - 8E(y^2) - R_{xy} = \dots$

⑦ $C_{vw} = R_{vw} - E(v)E(w) = \dots$

⑧ $\rho_{vw} = \frac{C_{vw}}{\sigma_w \sigma_v}$

(14)

So joint characteristic function??

$$\Phi_{xy}(\omega_1, \omega_2) = \iint_{-\infty}^{\infty} f_{xy}(x, y) e^{j\omega_1 x + j\omega_2 y} dx dy$$

$$f_{xy}(x, y) = \frac{1}{(2\pi)^2} \iint_{-\infty}^{\infty} \Phi_{xy}(\omega_1, \omega_2) e^{-j\omega_1 x - j\omega_2 y} d\omega_1 d\omega_2$$

- marginal characteristic function

$$1) \Phi_x(\omega_1) = \Phi_{xy}(\omega_1, 0) \quad 2) \Phi_y(\omega_2) = \Phi_{xy}(0, \omega_2)$$

$$m_{nk} = (-1)^{n+k} \frac{\partial^{n+k} \Phi_{xy}(\omega_1, \omega_2)}{\partial \omega_1^n \partial \omega_2^k} \bigg|_{\omega_1=0, \omega_2=0}$$

joint moment

• Example 2.0 two random variable have joint charac.

$$\Phi_{xy} = e^{(-2\omega_1^2 - 8\omega_2^2)}$$

find $\bar{x}, \bar{y}, R_{xy}, C_{xy}$

$$\bar{x} = E(x) = m_{10} = -j \frac{\partial \Phi_{xy}(\omega_1, \omega_2)}{\partial \omega_1} \bigg|_{\omega_1=0, \omega_2=0}$$

$$= j(-4\omega_1) e^{(-2\omega_1^2 - 8\omega_2^2)} \bigg|_{\substack{\omega_1=0 \\ \omega_2=0}} = 0$$

$$\bar{y} = E(y) = m_{01} = -j \frac{\partial \Phi_{xy}(\omega_1, \omega_2)}{\partial \omega_2} \bigg|_{\substack{\omega_1=0 \\ \omega_2=0}} = 0$$

• $R_{xy} = 0$ then $C_{xy} = 0$ mean x, y uncorr

(15)

for joint Gaussian Random variable ::

- two Random variable ::

Two random variables X and y are said to be joint Gaussian if their joint density function is of the form ::

$$f_{xy}(x,y) = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}}$$

$$\times \exp \left[-\frac{1}{2(1-\rho^2)} \left[\frac{(x-\bar{x})^2}{\sigma_x^2} - \frac{2\rho(x-\bar{x})(y-\bar{y})}{\sigma_x\sigma_y} + \frac{(y-\bar{y})^2}{\sigma_y^2} \right] \right]$$

- the maximum value when $\bar{x}=x$, $y=\bar{y}$

$$f_{xy}(x,y) = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}}$$

- if x, y uncorrelated ($\rho=0$) then

$$f_{xy}(x,y) = \frac{1}{2\pi\sigma_x\sigma_y} \exp \left[-\frac{1}{2} \left(\frac{(x-\bar{x})^2}{\sigma_x^2} + \frac{(y-\bar{y})^2}{\sigma_y^2} \right) \right]$$

- if y, x are uncorrelated and indep. then

$$f_x(x) = \frac{1}{\sqrt{2\pi\sigma_x^2}} \exp \left(-\frac{(x-\bar{x})^2}{2\sigma_x^2} \right)$$

(16.1)

$$f_y(y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} \exp\left(-\frac{(y-\bar{y})^2}{2\sigma_y^2}\right)$$

Example 2: Gaussian Random variables, consider random variable y_1 and y_2 related to random variables x and y by relation

$$y_1 = x \cos(\theta) + y \sin(\theta)$$

$$y_2 = -x \sin(\theta) + y \cos(\theta)$$

if $\bar{y}, \bar{x}$ are the mean of x, y , the means of y_1 and

$$y_2 \text{ are } \bar{y}_1 = \bar{x} \cos(\theta) + \bar{y} \sin(\theta)$$

$$\bar{y}_2 = -\bar{x} \sin \theta + \bar{y} \cos \theta$$

Find the covariance of y_1 and y_2 (C_{y_1, y_2})

$$-C_{y_1, y_2} = E[(y_1 - \bar{y}_1)(y_2 - \bar{y}_2)]$$

$$= E[(x - \bar{x} \cos \theta + y - \bar{y} \sin \theta)(- (x - \bar{x}) \sin \theta + y - \bar{y} \cos \theta)]$$

$$= (\sigma_y^2 - \sigma_x^2) \sin \theta \cos \theta + C_{xy} [\cos^2 \theta - \sin^2 \theta]$$

$$= (\sigma_y^2 - \sigma_x^2) \frac{1}{2} \sin 2\theta + C_{xy} \cos 2\theta$$

And note

$$E(x - \bar{x})^2 = \sigma_x^2$$

$$C_{xy} = E(x - \bar{x})(y - \bar{y})$$

$$E(y - \bar{y})^2 = \sigma_y^2$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

(17)

Chapter 6

6.1 Random processes ...

- Random processes its definition a function of the possible outcomes s of an experiment, it becomes a function of both s and time.

$$X(t) = A \cos(\omega_0 t + \theta)$$

where

A, θ , or ω_0 may be random variables

6.2 First-order stationary processes

A random process is called stationary to order one if the first order density function does not change with a shift in time origin.

$$f_X(x_1, t_1) = f_X(x_1, t_1 + \Delta)$$

must be true for any t_1 and any real number Δ if $X(t)$ is to be a first-order stationary process

$$E(X(t)) = \bar{X} = \text{constant}$$

$$E(X(t)) = E(A \cos \omega_0 t + \theta) = \bar{X} = \text{constant}$$

for Second order and wide-sense stationarity

A process is called stationary to order two if its second order density function statistics $\therefore$

$$f_X(x_1, x_2; t_1, t_2) = f_X(x_1, x_2; t_1 + \Delta, t_2 + \Delta)$$

• note A second-order process is also first-order because the second order density function determines the lower, first order density

$$E\{x(t_1, t_2)\} = E[x(t_1) \cdot x(t_2)] \quad \text{"general"}$$

- Autocorrelation Function $\therefore$

$$R_{xx}(t_1, t_2) = E[x(t_1) \cdot x(t_2)]$$

where $\tau = t_2 - t_1$ then

$$R_{xx}(t_1, t_1 + \tau) = E\{x(t_1) \cdot x(t_1 + \tau)\} = R_{xx}(\tau)$$

- properties for autocorr $\therefore$

ex. 1.1.1

$$1) |R_{xx}(\tau)| \leq R_{xx}(0)$$

$$2) R_{xx}(-\tau) = R_{xx}(\tau) \Rightarrow \text{has even symmetry}$$

$$3) R_{xx}(0) = E(x^2(t)) \text{ "power"} = m_2$$

$$4) \lim_{|\tau| \rightarrow \infty} R_{xx}(\tau) = \bar{x}^2 = (m_1)^2$$

6.01 Cross-Correlation function "R_{xy}"

Two Random variable processes $x(t)$ & $y(t)$ we say they are jointly wide-sense stationary and there cross correlation defined by

$$R_{xy}(t_1, t_2) = E[x(t_1) \cdot y(t_2)]$$

When $\tau = t_2 - t_1$ then

$$R_{xy}(t_1, t_1 + \tau) = E[x(t_1) \cdot y(t_1 + \tau)]$$

→ if $R_{xy}(t, t + \tau) = 0$ then x, y are called orthogonal process

→ if $R_{xy}(t, t + \tau) = E[x(t)] \cdot E[y(t + \tau)]$ then x, y are statistically independent

→ if $x(t), y(t)$ are wide sense stationary then $R_{xy}(\tau) = \bar{x}\bar{y}$

- properties of cross-correlation ∴

$$1) R_{xy}(\tau) = R_{xy}(-\tau)$$

$$2) |R_{xy}(\tau)| \leq \sqrt{R_{xx}(0) \cdot R_{yy}(0)}$$

$$3) R_{xy}(\tau) = \frac{1}{2} [R_{xx}(0) + R_{yy}(0)]$$

• wide-sense stationary ∴ "WSS"

$$1. E(x(t)) = \bar{x} = \text{constant}$$

$$2. E[x(t) \cdot x(t+\tau)] = R_{xx}(\tau)$$

إذا لم يتغير المتوسط يكون

Random process is wide-sense (WSS)

Example ∴ For the random process $x(t) = A \cos(\omega_0 t + \theta)$ where A and ω_0 are constants and θ is uniformly distributed random variable on the interval $(0, 2\pi)$ find if it wide sense stationary or not?

- الشرط الأول -

$$1) E(x(t)) = \int_0^{2\pi} A \cos(\omega_0 t + \theta) \cdot \frac{1}{2\pi} \cdot d\theta$$

uniform = $\frac{1}{b-a}$

= zero = constant

وهذا يعني أنه ثابت

(21)

2) الشرط الثاني

$$R_{xx}(t, t+\tau) = E(A \cos(\omega_0 t + \theta) * A \cos(\omega_0 t + \omega_0 \tau + \theta))$$

تسليم المتغيرات

$$= \frac{A^2}{2} E(\cos \omega_0 \tau + \cos(2\omega_0 t + \omega_0 \tau + 2\theta))$$

$$\frac{A^2}{2} \int (\cos(\omega_0 \tau) + \cos(2\omega_0 t + \omega_0 \tau + 2\theta)) \cdot \frac{1}{b-a} \cdot d\theta$$

$$\frac{A^2}{2} \cdot \frac{1}{2\pi} \left[\int_0^{2\pi} \cos(\omega_0 \tau) \cdot d\theta + \int_0^{2\pi} \cos(2\omega_0 t + \omega_0 \tau + 2\theta) d\theta \right]$$

zero

$$\frac{A^2}{2} \times \frac{1}{2\pi} \cos \omega_0 \tau \cdot \theta \Big|_0^{2\pi} \Rightarrow \frac{A^2}{2} \cos \omega_0 \tau$$

بالتالي τ ثابت
تبقى الشرط الثاني

Example :: if $R_{xx}(\tau) = 25 + \frac{4}{1 + 6\tau^2}$ find σ_x^2

$$\sigma_x^2 = m_2 - (m_1)^2 \Rightarrow 29 - 25 = 4$$

$$-(m_1)^2 = E(x(t))^2 = \lim_{|\tau| \rightarrow \infty} R_{xx}(\tau)$$

$$= 25 + \frac{4}{1+\infty} = 25$$

$$m_2 = E(x(t)^2) = R_{xx}(0) = 29$$

Example 3: Consider random process $X(t) = A \cos(\omega t + \phi)$.
 $X(t) = A \cos(\omega t + \phi) + 3$ if A and ϕ is indep R.V
 Uniform distribution $A (0 \rightarrow 2)$, $\phi \rightarrow (-\pi, \pi)$ Find

1) mean value of process

$$E(X(t)) = E(A \cos(\omega t + \phi) + 3)$$

$$= \int_{-\pi}^{\pi} \int_0^2 A \cos(\omega t + \phi) + 3 \, dA \, d\phi \quad \frac{1}{2\pi} \cdot \frac{1}{2}$$

$$= \frac{1}{4\pi} \int_{-\pi}^{\pi} \frac{A^2}{2} \cos(\omega t + \phi) + 3 \cdot d\phi$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos(\omega t + \phi) + 3 \, d\phi$$

$$= \frac{1}{2\pi} \left[\underbrace{\sin(\omega t + \phi)}_{\text{zero}} + 3\phi \right] \Big|_{-\pi}^{\pi}$$

$$3 \times (\pi - (-\pi)) \cdot \frac{1}{2\pi} \Rightarrow 3 = \text{constant}$$

$$- R_{xx}(\tau) = E(x(t) \cdot x(t+\tau))$$

$$= E((A \cos(\omega t + \phi) + 3) \times A \cos(\omega t + \omega \tau + \phi) + 3)$$

$$\Rightarrow \frac{A^2}{2} \cos(\omega \tau) + 27$$

بما أن A عشوائي، والتوزيع العشوائي

Example

Let two random processes $x(t)$ and $y(t)$ be defined by

$$x(t) = A \cos \omega_0 t + B \sin \omega_0 t$$

$$y(t) = B \cos \omega_0 t - A \sin \omega_0 t$$

where A, B are random variable and ω_0 constant if A, B uncorrelated, zero mean random variable with same variance. Find $R_{xy}(t, t+\tau)$ and show that $x(t), y(t)$ are wide sense stationary ??

$$R_{xy}(t, t+\tau) = E(x(t) \cdot y(t+\tau))$$

$$E[AB \cos \omega_0 t \cos(\omega_0 t + \omega_0 \tau) + B^2 \sin \omega_0 t \cos(\omega_0 t + \omega_0 \tau) - A^2 \cos \omega_0 t \sin(\omega_0 t + \omega_0 \tau) - AB \sin \omega_0 t \sin(\omega_0 t + \omega_0 \tau)]$$

$$= E[AB] \cos(2\omega_0 t + \omega_0 \tau) + E[B^2] \sin \omega_0 t \cos(\omega_0 t + \omega_0 \tau) - E[A^2] \cos \omega_0 t \sin(\omega_0 t + \omega_0 \tau)$$

where $E[AB] = 0$ and $E[A^2] = E[B^2] = \sigma^2$

$$\text{So } R_{xy}(t, t+\tau) = -\sigma^2 \sin \omega_0 \tau$$

Thus $x(t)$ and $y(t)$ are jointly wide sense stationary
 $R_{xy}(t, t+\tau)$ depends only on τ

Def Covariance Functions :

- the autocovariance function is defined by

$$C_{xx}(t, t+\tau) = E[(x(t) - E[x(t)])(x(t+\tau) - E[x(t+\tau)])]$$

also can be write in the form

$$C_{xx}(t, t+\tau) = R_{xx}(t, t+\tau) - E(x(t))E(x(t+\tau))$$

- the cross-covariance function for two random processes $x(t)$ and $y(t)$ is defined by

$$C_{xy}(t, t+\tau) = R_{xy}(t, t+\tau) - E(x(t))E(y(t+\tau))$$

For processes that are at least jointly wide sense stationary then :

$$C_{xx}(\tau) = R_{xx}(\tau) - \bar{x}^2$$

and

$$C_{xy}(\tau) = R_{xy}(\tau) - \bar{x}\bar{y}$$

* Time Average and ergodic process :-

Time Average mathematically give by

$$\begin{aligned} - A[x(t)] &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x^2(t) dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x^2(t) dt \end{aligned}$$

and the out correlation function

$$- R_{xx}(t, t+T) = A[x(t) \cdot x(t+T)]$$

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) \cdot x(t+T) dt$$

$$\text{if } A(x(t)) = E(x(t))$$

and

$$R_{xx}(t, t+T) = R_{xx}(t, t+T)$$

Average time

The process called ergodic process

if $x(t)$ $y(t)$ zero mean and orthogonal and $R_{yy}(\tau) = \cos(2\pi 10^6 \tau)$ let $w(t) = x(t) + 2y(t)$.
 the $R_{ww}(\tau) = ?$ $R_{xx}(\tau) = e^{-|\tau|}$

$$\begin{aligned} \text{Sol: } R_{ww}(\tau) &= E[w(t_1)w(t_1+\tau)] = \\ &= E[x(t_1) + 2y(t_1)][x(t_1+\tau) + 2y(t_1+\tau)] \\ &= E[x(t_1)x(t_1+\tau) + 2x(t_1)y(t_1+\tau) \\ &\quad + E[2y(t_1)x(t_1+\tau) + 4y(t_1)y(t_1+\tau)] \\ &= E[x(t_1)x(t_1+\tau)] + E[2x(t_1)y(t_1+\tau)] \\ &\quad + E[2y(t_1)x(t_1+\tau)] + E[4y(t_1)y(t_1+\tau)] \\ &= R_{xx}(\tau) + 2R_{xy}(\tau) + 2R_{yx}(\tau) + 4R_{yy}(\tau) \\ &= R_{xx}(\tau) + R_{yy}(\tau) \end{aligned}$$

$$\therefore R_{xy}(\tau) = R_{yx}(\tau) = 0 \quad \text{orthogonal}$$

$$R_{ww}(\tau) = e^{-|\tau|} + 4 \cos(2\pi 10^6 \tau).$$

Consider Random process $x(t) = A \cos(\omega t + \phi)$
 and ϕ R.V Uniform distrib $[-\pi, \pi]$
 if $y(t) = z(t) \cdot x(t)$ where $z(t)$ is WSS and
 $R_{zz} = e^{-2|\tau|}$ and find R_{xx}, R_{yy}
 and $y(t)$ WSS or not??

$$R_{xx} = E(x(t) \cdot x(t+\tau)) = E(A \cos(\omega t + \phi) \cdot A \cos(\omega(t+\tau) + \phi))$$

نفسه مرقب

$$E\left(\frac{A^2}{2} (\cos \omega \tau) + \cos(2\omega t + 2\phi + \omega \tau)\right)$$

$$\frac{1}{2\pi} \cdot \frac{A^2}{2} \int_{-\pi}^{\pi} (\cos(\omega \tau) + \cos(2\omega t + 2\phi + \omega \tau)) d\phi$$

$$\therefore R_{xx}(\tau) = E[y(t_1) y(t_1 + \tau)] \quad y(t_1) = z(t_1) x(t_1)$$

$$y(t_1 + \tau) = z(t_1 + \tau) x(t_1 + \tau)$$

$$R_{yy}(\tau) = E[z(t_1) \cdot x(t_1) \cdot z(t_1 + \tau) x(t_1 + \tau)]$$

$$= E[z(t_1) z(t_1 + \tau) x(t_1) x(t_1 + \tau)]$$

نوع ال E على الخرج لان $z(t), x(t)$ indep.

$$R_{yy}(\tau) = E(z(t_1) z(t_1 + \tau)) \cdot E(x(t_1) x(t_1 + \tau))$$

$$= R_{zz}(\tau) \cdot R_{xx}(\tau)$$

$$= \frac{A^2}{2} \cos(\omega \tau) e^{-2|\tau|}$$

(28)

③ ① $E(y(t_1)) = E(x(t_1)) E(z(t_1)) = \text{constant}$ } is ~~not~~ WSS

$$\textcircled{2} R_{yy}(\tau) = \frac{A^2}{2} \cos(\omega \tau) e^{-2|\tau|}$$